

SunoBolo — Voice-Based Inventory Management Using Conversational Speech

Milestone 1.3 · Technical Design & Architecture

Deliverable: Technical Document + Architecture Diagram

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Prepared by: _____

Project Context

SunoBolo is a voice-first inventory system for small Indian retail shops.

Every voice inventory product on the market works as a **command interface**: you press a button and issue a structured instruction — *"add twenty kilo rice."* That is still data entry, performed with your mouth instead of your thumbs. SunoBolo inverts this. The phone sits on the shop counter and **listens to the shop**, extracting stock movements from the ordinary conversation between owner and customer — sentences that were never addressed to the app at all. A kirana shop already says every transaction out loud, usually twice: once when the customer asks and once when the owner confirms. That speech is a complete, free transaction log that currently evaporates into the air. **SunoBolo captures it.**

Because passive capture is less accurate than dictated commands, nothing it extracts is written to stock automatically. Every extraction becomes a **proposal** the owner approves in bulk, and a conventional **Command Mode** mic button remains available as the reliable path. The novel capability is genuinely novel; the failure mode is an ordinary voice inventory app.

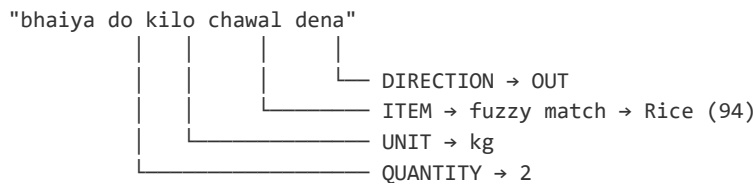
Milestone 1 deliverables

- **1.1** — Requirements Gathering & Analysis
- **1.2** — Functional Requirements
- **1.3** — Technical Design & Architecture (*this document*)
- **1.4** — UI/UX Design & Prototype

Technical Design & Architecture

1. Design Thesis

Inventory speech is a **closed-world problem**. A shop's utterances are unbounded in form, but the answer space is tiny: roughly 50 items, 10 units, 4 actions. The system therefore never attempts to *understand sentences*. It scans a noisy transcript for four known slots, each resolved against a finite lexicon:



Three consequences follow, and they define the architecture:

1. **No model is trained.** The mapping a model would have to learn is small enough to write down as data. Recognition is off-the-shelf; understanding is deterministic and testable.

2. **One fuzzy matcher absorbs two different problems.** Speech-recognition error

(*chaawal*, *chowal*) and language variation (*rice*, चावल, *biyyam*) are both just

near-misses against an alias list. A single `rapidfuzz` call handles both — so the system

needs neither a spell-checker nor a translator. **This is the whole trick.**

3. **Because passive capture is unreliable, nothing it produces may auto-commit.** The parse

step is read-only by construction. This converts a probabilistic component into a safe one:

the worst outcome of a bad extraction is one dismissed card, never a corrupted ledger.

2. Technology Stack

Layer	Choice	Rationale
Speech-to-text	Web Speech API (<code>webkitSpeechRecognition</code> , <code>continuous = true</code>)	Free, no key, no model download, sub-second; supports hi-IN , te-IN , en-IN . Continuous mode is what makes passive capture possible at zero cost
Text-to-speech	Web Speech Synthesis	Native per-locale voices, zero cost
Frontend	Single HTML file, vanilla JS, plain	No framework, no build step, no

	CSS	<code>node_modules</code> . The entire UI is one readable file
Backend	FastAPI (Python 3.11)	Automatic OpenAPI docs; minimal boilerplate
Database	<code>`sqlite3`</code> from the standard library (dev) / PostgreSQL (prod)	No ORM. The schema is four tables and the queries are short enough to read directly
NLU	<code>`rapidfuzz`</code> + lexicon tables	Deterministic, testable, inspectable, fast
Hosting	Render — one service, FastAPI serves the HTML	Single deploy, single origin, no CORS

Why recognition lives in the browser. Server-side transcription (Whisper) would add a 250 MB model, 5–15 s of CPU latency per utterance on free hosting, and a hard RAM floor — and continuous listening would make that cost permanent. The browser recogniser returns text in under a second at zero cost, with **no audio ever leaving the device**, which is also what makes the privacy position defensible.

Why no frontend framework. The app has four screens and one list that updates. React's value is managing complex state; there isn't any here. Dropping it removes a build step, a lockfile, 1,400 dependency files, and roughly 300 lines of code.

3. Architecture Diagram

```

flowchart TB
    subgraph Counter["📱 Phone on the shop counter"]
        CONV["🗣️ Owner ↔ Customer conversation<br/>never addressed to the app"]
        WSA["Web Speech API · continuous = true<br/>auto-restarts on silence<br/>NO AUDIO STORED OR UPLOADED"]
        SEG["Rolling segments"]
        CMD["🖱️ Command Mode button<br/>deliberate single entry"]
        UI["UI · one HTML file<br/>Dashboard · Pending tray · Alerts · Item detail"]
        TTS["Speech Synthesis<br/>spoken reply in user's language"]
        CONV --> WSA --> SEG --> UI
        CMD --> WSA
        UI --> TTS
    end

    subgraph API["⚙️ FastAPI on Render"]
        PARSE["POST /parse · READ-ONLY, never writes"]
        TXN["POST /commit · the only write path"]
        STOCK["GET /stock"]
        ALERT["GET /alerts"]
    end

    subgraph NLU["NLU – rules + fuzzy · no ML, no training"]
        direction TB
        N1["1 · Normalise"]
    end

```

```

N2["2 · Direction → IN / OUT / QUERY<br/>counter default = OUT"]
N3["3 · Quantity<br/>digits + number-words<br/>bees=20 · teen sau=300 ·
dhai=2.5"]
N4["4 · Unit → base<br/>kilo/bori/dozen/peti/tin"]
N5["5 · Item<br/>rapidfuzz token_set_ratio<br/>vs multilingual alias table"]
N6["6 · Confidence gate<br/>≥85 accept · 60-85 confirm · &lt;60 DISCARD"]
N1 --> N2 --> N3 --> N4 --> N5 --> N6
end
PARSE --> NLU
end

subgraph DB["🗄️ SQLite / PostgreSQL"]
  T2[("items · aliases[] · base_unit<br/>reorder_level · target_level")]
  T3[("item_units · per-item pack sizes<br/>1 bori rice = 50 kg")]
  T4[("transactions – APPEND ONLY<br/>qty_base · transcript · confidence")]
end

UI -- "segment text + lang" --> PARSE
PARSE -. "PROPOSALS – nothing written" .-> TRAY
TRAY["📋 Pending tray<br/>bulk accept · one-tap correct"]
TRAY -- "owner approved" --> TXN
UI --> STOCK
UI --> ALERT
N5 -- "alias lookup" --> T2
TXN -- "unit → base" --> T3
TXN --> T4
STOCK -- "SUM(qty_base) GROUP BY item" --> T4
ALERT -- "stock ≤ reorder_level" --> T4

style NLU fill:#fff4e6,stroke:#f59e0b,stroke-width:2px
style Counter fill:#eef6ff,stroke:#3b82f6,stroke-width:2px
style DB fill:#f0fdf4,stroke:#22c55e,stroke-width:2px
style TRAY fill:#fef9c3,stroke:#ca8a04,stroke-width:2px

```

The request path for one overheard sale:

```

conversation → continuous recognition → segment text
               → POST /parse {text, lang}           ← READ-ONLY
               → proposals [{item, qty, unit, dir, confidence}]
               → pending tray (nothing written yet)
               → owner taps accept → POST /commit    ← the only write
               → new total returned → spoken + displayed

```

4. Data Model

```

items (
  id          INTEGER PRIMARY KEY,
  name        TEXT NOT NULL,          -- canonical display name
  aliases     TEXT NOT NULL,          -- JSON array: all langs + ASR misspellings
  base_unit   TEXT NOT NULL,          -- 'kg' | 'pc' | 'litre'
  reorder_level REAL DEFAULT 0,       -- in base_unit
  target_level REAL DEFAULT 0,       -- restock target, drives suggestions
  last_price  REAL
);

item_units (                          -- per-item pack sizes

```

```

    item_id    INTEGER REFERENCES items(id),
    unit       TEXT NOT NULL,          -- 'bori', 'carton', 'tray'
    factor     REAL NOT NULL           -- 1 unit = factor × base_unit
);

transactions (                        -- append-only ledger
    id        INTEGER PRIMARY KEY,
    item_id   INTEGER REFERENCES items(id),
    qty_spoken REAL NOT NULL,         -- 2
    unit_spoken TEXT NOT NULL,        -- 'bori'
    qty_base   REAL NOT NULL,         -- +100, or negative for 'out'
    price     REAL,
    transcript TEXT,                  -- exact words heard – audit trail
    lang       TEXT,
    confidence REAL,
    source     TEXT,                  -- 'counter' | 'command' | 'manual' | 'undo'
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

shops (id, name, pin_hash, lang);

```

4.1 Stock is derived, never stored

```
SELECT item_id, SUM(qty_base) AS stock FROM transactions GROUP BY item_id;
```

One decision delivers four requirements at once:

- **Transaction history** — it *is* the storage format, not a parallel log
- **Undo** — insert a reversing row; nothing is ever deleted
- **Audit trail** — every number traces to the sentence that produced it
- **Correction safety** — a mis-parse is amendable, never destructive

Ceiling: stock is recomputed by aggregate on every read. Correct and trivially consistent at

shop scale. A cached column with a trigger is warranted only past ~100k transactions.

4.2 Unit conversion — two layers

Global units, universal and hard-coded: *gram* → *kg* ×0.001, *quintal* → *kg* ×100,

dozen → *pc* ×12, *tray* → *pc* ×30, *ml* → *litre* ×0.001.

Per-item pack sizes live in *item_units*, because they are genuinely item-specific:

Item	Unit	Factor
Rice	bori	50 kg
Sugar	bag	25 kg
Maggi	carton	96 pc
Sunflower oil	tin	15 litre

Resolution order: per-item → global → item's base unit. Every transaction stores **both** the

spoken unit and the base-unit value, so the owner speaks in *bori*, the database reasons in

kilograms, and the UI displays either.

5. The NLU Pipeline

Input: transcript plus a language hint. Output: zero or more structured movements with confidence scores. A pure function over the item catalogue — therefore fully unit-testable.

Step 1 — Normalise. Lowercase, strip punctuation, collapse whitespace. Devanagari and Telugu pass through unchanged; the matcher is script-agnostic.

Step 2 — Direction.

Intent	Keywords (hi / te / en)
IN	aaya, aa gaya, liya, kharida, mangwaya · vacchindi, konnanu · came, bought, received, add
OUT	dena, de do, becha, diya, nikala, chahiye · kavali, ammanu, ichanu · sold, give, need
QUERY_ITEM	kitna, kitne, bacha, batao · enta, unnayi, cheppu · how much, how many
QUERY_LOW	khatam, kam, mangwana · takkuva · running low, finished, order

Priority QUERY_LOW → QUERY_ITEM → IN → OUT. **In Counter Mode the default is OUT**, because counter conversation is overwhelmingly selling; in Command Mode the default is IN, because deliberate entries are usually deliveries.

Step 3 — Quantity. Digits taken directly. Number words resolved from a map covering 1–100 plus *sau/hazaar/nooru/veyyi*, with multiplier composition ("*teen sau*" → 3 × 100) and the fractional trade words *dhai* 2.5, *derh* 1.5, *sawa* 1.25, *paune* 0.75. Absent → 1.

Step 4 — Unit. Matched against the combined unit lexicon in all languages, singular and plural. Absent → the item's base unit.

Step 5 — Item. The load-bearing step. `rapidfuzz.process.extract` with `token_set_ratio` against every alias of every item. One lookup simultaneously absorbs:

- **ASR noise** — *chaawal, chowal, chawl* → Rice
- **Language variation** — *rice, chawal, चावल, biyyam, బియ్యం* → Rice
- **Partial phrases** — *sunflower oil, tel, nune* → Sunflower Oil

This is precisely why there is no translation layer and no spell-correction layer: both are the same problem, and one similarity search solves it.

Step 6 — Multi-item split. The segment is split on conjunctions (*aur, and, mariyu*)

and each fragment parsed independently, so one sentence can yield several movements.

Step 7 — Confidence gate. Scores map to the behaviour table in FR-4.3 of the Functional Requirements document. In Counter Mode

anything below 60 is **discarded silently** rather than shown — the cost of a false proposal is the owner's attention, and attention is the scarce resource being protected.

6. API Surface

Method	Route	Purpose
POST	/parse	{text, lang, mode} → proposals + confidence. Read-only
POST	/commit	Write one approved movement; returns the new total
POST	/commit/bulk	Write all approved proposals in one call
POST	/undo/{id}	Insert the reversing row
GET	/stock	All items with derived quantity and status
GET	/alerts	Low-stock list with reorder suggestions
GET	/item/{id}	Detail with pack sizes and recent history
GET	/export.csv	Full ledger export (backup)
POST	/login	Shop name + PIN → session token

7. Non-Functional Design

Privacy. Recognition is browser-local; no audio is recorded, stored or uploaded. Only the extracted values and the resulting transcript reach the server. Counter Mode displays an unmistakable listening indicator whenever it is active.

Authentication. Shop name plus a 4-digit PIN stored as a salted SHA-256 hash; session token in `localStorage`. A 4-digit PIN is the right control for a single-operator shop app — memorable, and no keyboard.

Ceiling: move to bcrypt and short-lived JWTs before any real multi-tenant deployment.

Logging. Every parse is logged with its transcript, chosen item, score, and whether the owner corrected it. This is both an operational log and the feedback signal for extending the alias lexicon — the corrections name exactly which aliases are missing.

Degradation. Without the Web Speech API the transcript field becomes a text input feeding the identical `/parse` endpoint. The fallback is not a separate code path; it is the manual-entry requirement (FR-4.6 of the Functional Requirements document).

Accessibility. 48 px minimum touch targets; status conveyed by colour *and* icon *and* text; system font stack with Devanagari and Telugu coverage; one-handed operation.

8. Quality Assurance

Layer	Approach
NLU	<code>test_parse.py</code> — a table of utterances across all three languages asserted against expected extractions. The parser is a pure function, so the suite runs in under a second
Unit conversion	Assertions on the per-item and global factor chain, including <code>"2 bori rice = 100 kg"</code>
API	FastAPI <code>TestClient</code> over the commit-then-read path
End-to-end	Scripted manual run on a real phone against the deployed URL, including a performed counter conversation

The parser test table doubles as the specification of supported phrasing, and grows directly from the correction log above.

9. Deployment

One Render web service. FastAPI serves the static HTML and the API from the same origin — no CORS, no separate frontend deploy, no build step. PostgreSQL is a managed Render add-on; `DATABASE_URL` switches between it and local SQLite.

The Web Speech API requires HTTPS, which Render provides.

10. Code Budget

File	Lines	Responsibility
<code>parse.py</code>	~110	Lexicon, slot scanner, fuzzy matching, confidence — the only clever file
<code>app.py</code>	~130	FastAPI routes, <code>sqlite3</code> access, seed data
<code>index.html</code>	~190	Entire UI + continuous listener, no framework
<code>test_parse.py</code>	~25	Parser assertions
Total	~455	No build step, no lockfile, no <code>node_modules</code>

Appendix — Honest Assessment of Risk

The passive-capture approach is deliberately more ambitious than command-mode voice entry, and it is weaker in one specific respect: **extraction accuracy from unstructured conversation will be materially lower than from deliberate commands** — realistically 50–65% against roughly 90%.

This is designed around rather than hidden:

- Counter Mode **proposes**; it never commits. A wrong extraction costs one dismissive tap.
- Below-threshold extractions are discarded rather than shown, so the tray stays trustworthy.
- **Command Mode remains available at all times** as the reliable path. The passive feature is

the differentiator; the command path is the floor the product cannot fall through.

The result is a system whose novel capability is genuinely novel, and whose failure mode is an ordinary, well-understood voice inventory app.